

Applied Engineering & Robotics – Syllabus

School Year: 2025-2026

Instructor: Mr. Bombich

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Room: 70

Office Hours: Monday–Thursday, 2:45–3:15 pm

Prerequisites: None; basic math/science helpful.

Course Description

Dive into a year-long, hands-on journey to design, build, and program robots! In this project-based course, you'll master the Engineering Design Process (EDP) to create robots using CAD modeling, 3D printing, C++ programming for Arduino, and advanced shop tools like CNC machines, mills, and lathes. In Marking Period 1, you'll learn EDP, CAD, C++ coding, and build a base robot. In Marking Period 2, you'll add sensors, mechanisms, and new shop skills, plus start career prep with resume writing. In Marking Periods 3 and 4, you'll choose between solving real-world problems with EDP or designing FIRST Robotics competition mechanisms, wrapping up with mock interviews and a capstone project. Safety, teamwork, and ethical design are key throughout. Get ready to engineer solutions that make a difference!

Academic Credits

4th year related math, 3rd year science

College Credits

Pending approval, KVCC

Course Objectives

By the end of the year, you'll be able to:

- Apply the seven-step EDP (Ask, Research, Imagine, Plan, Create, Test, Improve) to solve engineering challenges.
- Create detailed 2D sketches, 3D CAD models, and technical drawings with precise measurements.
- Program Arduino microcontrollers using C++ to control motors, sensors, and mechanisms.
- Build and test prototypes, including 3D-printed parts, and analyze performance with data.
- Assemble electronic circuits with sensors (e.g., IR, ultrasonic, IMU) and troubleshoot issues.
- Practice shop safety and organization (5S, PPE, lockout/tagout, safety data sheets).

- Fabricate parts using CNC machines, mills, lathes, and hand tools with precision.
- Design and iterate robot mechanisms for functionality or FIRST Robotics competition.
- Test and refine designs using data, failure analysis (FMEA, RCA), and Bills of Materials (BOM).
- Collaborate in teams and deliver professional presentations.
- Build career skills through resume writing, mock interviews, and engineering career exploration.
- Evaluate the ethical, environmental, and societal impacts of your designs.

Materials & Tools

- Hardware: PC with Fusion 360, Arduino IDE; robot kit (chassis, motors, wheels), multimeter, soldering iron.
- Prototyping: 3D printer (PLA filament), tabletop CNC (polycarbonate sheets).
- Electronics: Arduino Nano RP2040 Connect, breadboards, jumper wires, sensors (TCRT5000 IR, MPU-6050 IMU, ultrasonic distance).
- Shop: CNC mill, manual mill, lathe, hand tools (e.g., calipers, wrenches).
- Safety: Safety glasses, gloves, ear protection, spark-resistant clothing (provided).
- Resources: Google Classroom for assignments; Autodesk Fusion 360 tutorials (autodesk.com/education); Arduino C++ guides (arduino.cc); safety data sheets (SDS).

Grading Breakdown

Category	Weight	Description
Participation, 5S, & Safety	40%	Daily engagement in lessons, labs, teamwork, journal entries, weekly shop cleanup, adherence to safety protocols.
Projects	40%	2–3 group projects per marking period (e.g., EDP, CAD, coding, mechanism builds) and final capstone project (EDP or FIRST), including reports and presentations.
Quizzes & Exams	20%	Unit quizzes; midterm and final exams (cumulative quiz content).

Unit & Pacing Overview

Marking Period 1: Foundations (Weeks 1–9)

Focus: Build foundational skills in EDP, CAD, C++ programming, and base robot construction.

Unit	Title	Duration	Key Topics
1	Engineering Design Process & Safety	3 weeks	EDP steps (Ask, Research, Imagine, Plan, Create, Test, Improve); problem-solving; shop safety (PPE, 5S, lockout/tagout); intro to engineering careers and ethics.
2	CAD Fundamentals	3 weeks	2D sketching, geometric constraints, 3D modeling in Fusion 360; technical drawings (multiview, dimensions); math for CAD (e.g., area, volume).
3	C++ Programming & Base Robot Build	3 weeks	C++ basics for Arduino (variables, loops, conditionals); motor control; breadboarding circuits; assemble base robot (chassis, motors, wiring).

Marking Period 2: Sensors, Mechanisms & Shop Skills (Weeks 10–18)

Focus: Add sensors, mechanisms, advanced shop skills, and career prep.

Unit	Title	Duration	Key Topics
4	Sensors & Advanced Programming	3 weeks	Advanced C++ (functions, sensor data processing); integrate sensors (IR, ultrasonic, IMU); troubleshoot circuits with multimeters.
5	Mechanisms & Shop Skills	3 weeks	Design simple mechanisms (e.g., levers, gears); CNC milling, manual mill, lathe operations; precision measurements; material properties.
6	Career Prep & Integration	3 weeks	Resume writing; combine sensors and mechanisms in robot builds; quality control basics; ethical design considerations.

Marking Period 3: Project Choice & Iteration (Weeks 19–27)

Focus: Choose between EDP projects or FIRST Robotics mechanisms, with iterative design.

Unit	Title	Duration	Key Topics
7	Project Design & Prototyping	4 weeks	Option 1: EDP project (real-world problem, e.g., sustainable device); Option 2: FIRST Robotics mechanism (e.g., arm, drivetrain); prototyping (3D printing, CNC); BOM creation.
8	Testing & Iteration	5 weeks	Test prototypes (EDP or FIRST); data analysis (performance metrics); failure analysis (FMEA, RCA); iterative redesign; teamwork documentation.

Marking Period 4: Capstone & Competition (Weeks 28–36)

Focus: Finalize projects, compete, and prepare for careers.

Unit	Title	Duration	Key Topics
9	Capstone Project & Competition	5 weeks	Finalize EDP project or FIRST Robotics mechanism; competition (robot or project showcase); data-driven reports; professional presentations; sustainability reflections.
10	Career Readiness & Wrap-Up	4 weeks	Mock interviews; career portfolio development; final capstone presentation; ethical and environmental impact analysis.

Weekly Structure

- Mon–Thu: Lesson (e.g., EDP, CAD, sensor coding, mill operation), lab activity (build, code, or test).
- Fri: 5S Fridays (shop cleanup, safety audits, starting Week 2) or project reviews (e.g., peer feedback on designs).
- Homework: Short videos (e.g., CNC tutorials), journal entries (e.g., EDP steps, career goals), or write-ups (e.g., mechanism ideas).

Classroom Policies

- Safety First: PPE mandatory in the shop; follow all rules. Violations = zero for the day.
- Attendance: Be present and active; unexcused absences affect your grade.
- Late Work: –10% per day (max 3 days); talk to me for extensions.

- Teamwork: Document your role in group projects; value diverse perspectives.
- Integrity: No plagiarism; credit all sources (e.g., code, research).
- Support: IEPs/504s supported; discuss accommodations privately.

Contact & Support

- Google Classroom: All assignments, rubrics, and resources.
- Email: Replies within 24 hours (weekdays).
- Open Lab: TBD